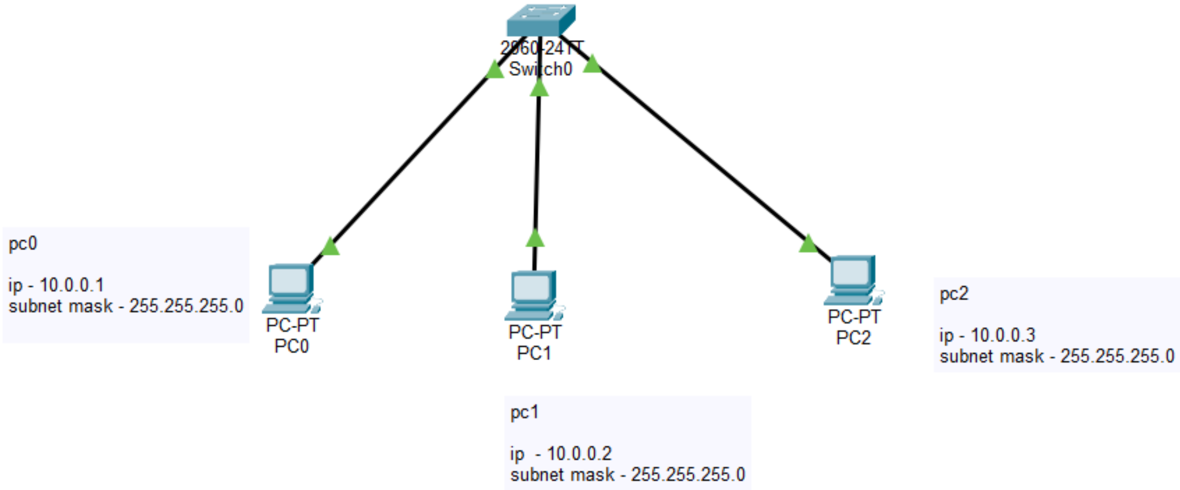
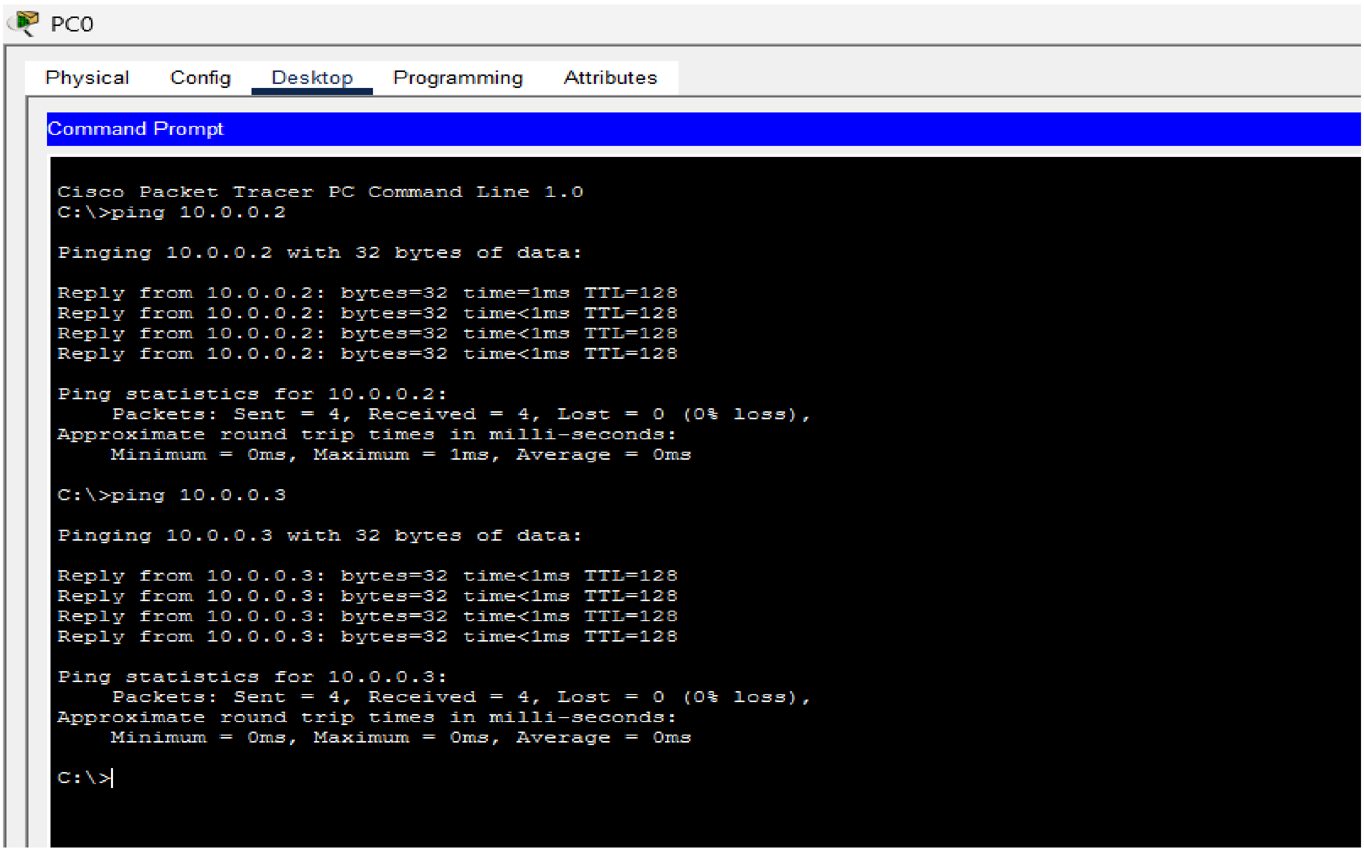
Tasks

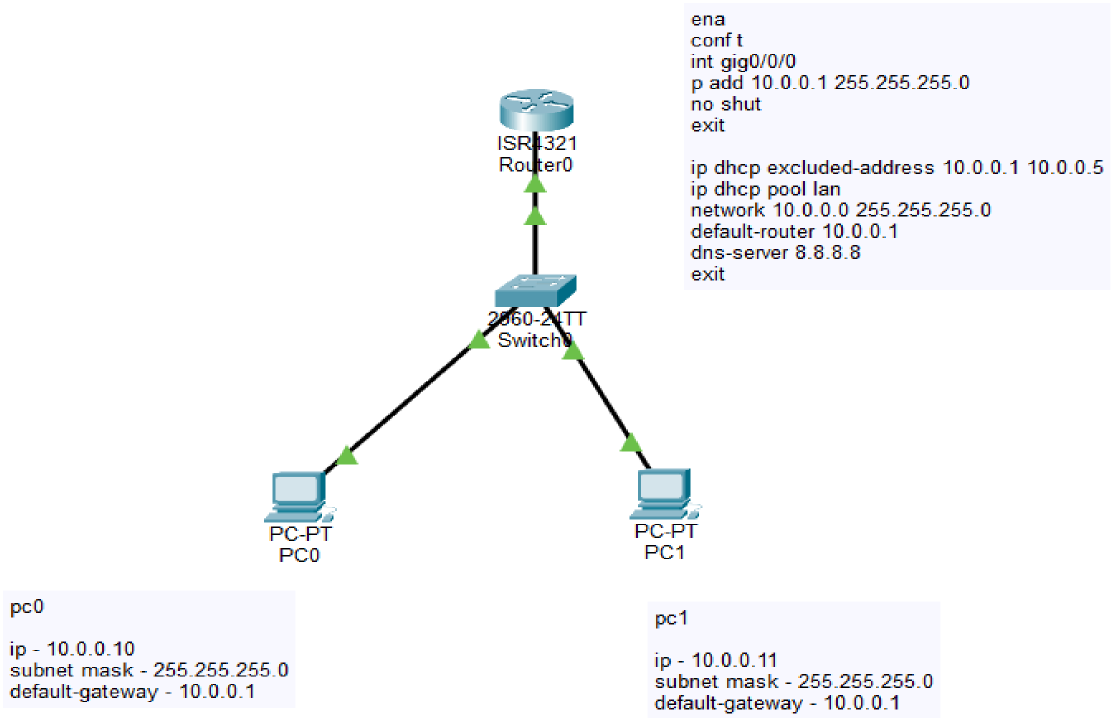
1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.



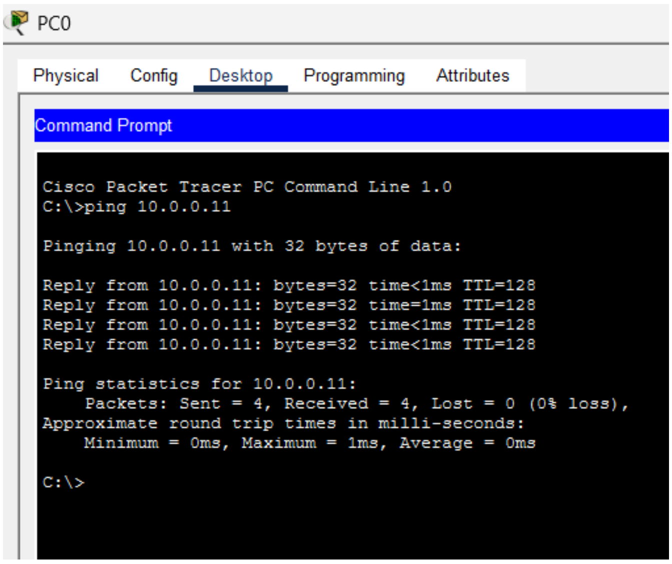
Result :-

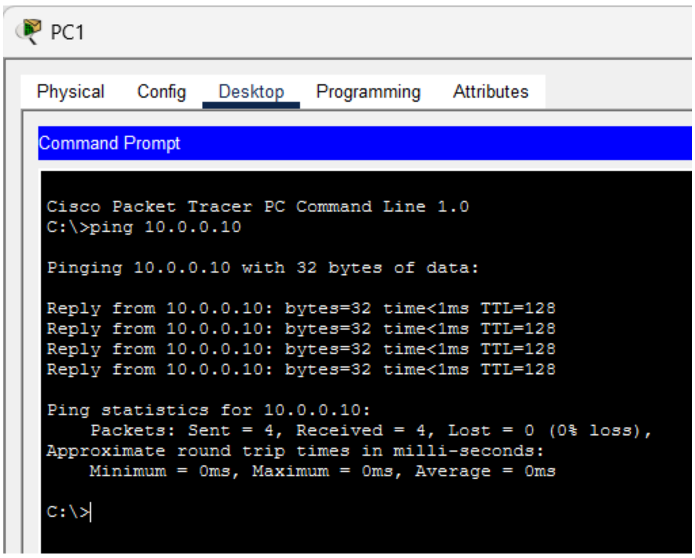


2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.



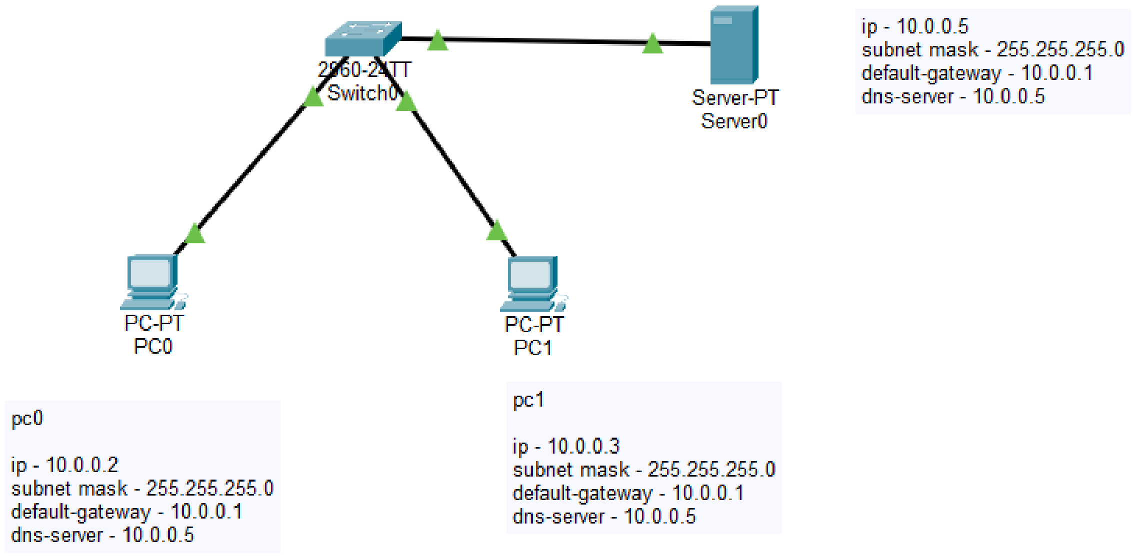
Result :-





3.Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Step 1:-



Step 2 :- Server0 → **Services → DNS**

- DNS Service = **ON**
- Table Entry
  - Name :- musicplayer.local
  - Address :- 10.0.0.5
  - Type :- A Record

Server0 ⟶ **Services** ⟶ **HTTP**

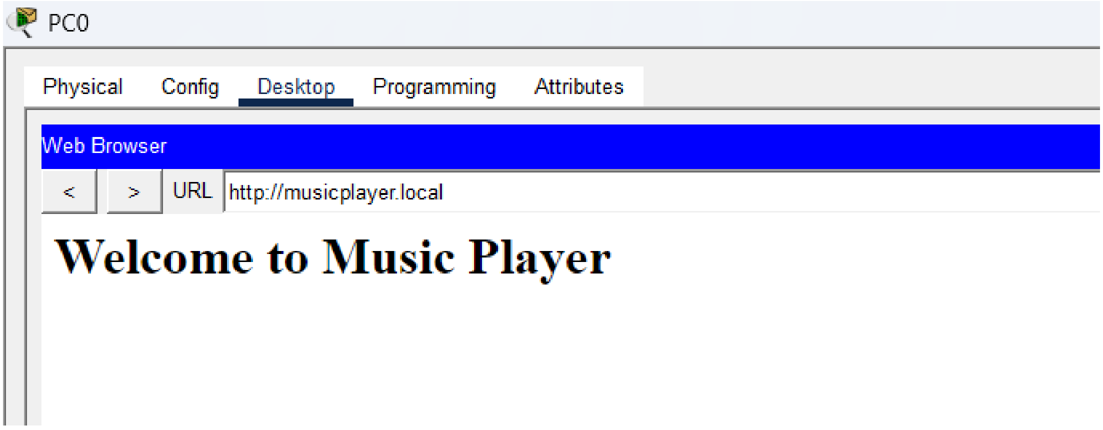
- HTTP = **ON**

Step 3:- pc0 ----- ping 10.0.0.5 (dns server)

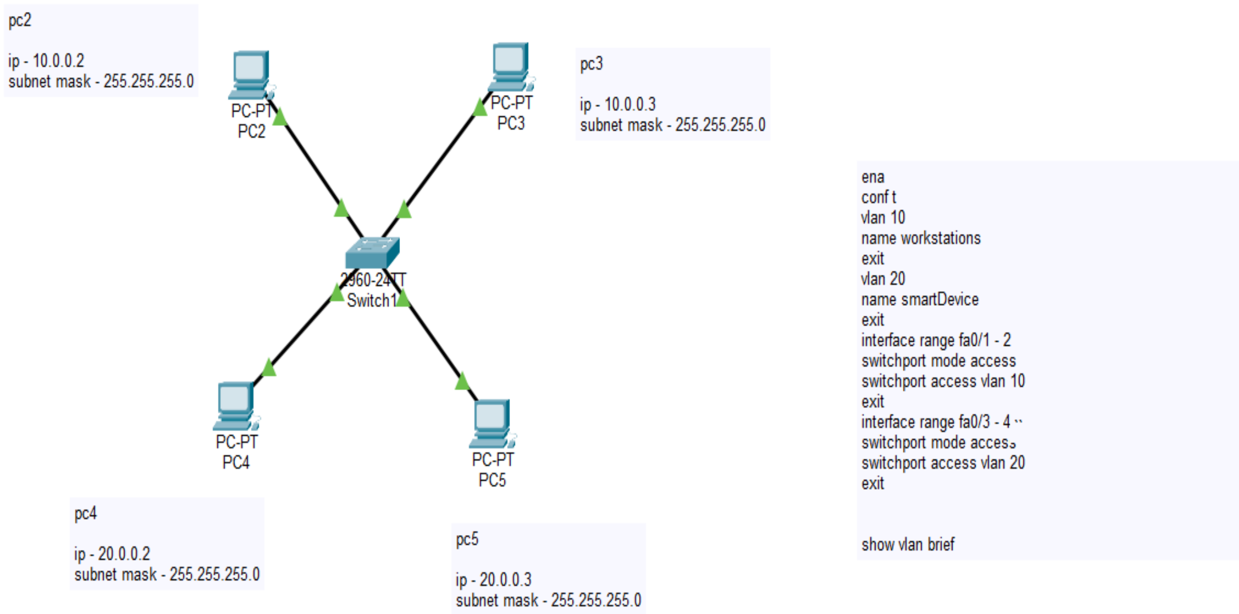
Step 4:- pc0 ----- ping musicplayer.local

Step 5:- pc0 ----- web browser ---- http://10.0.0.5 / http://musicplayer.local

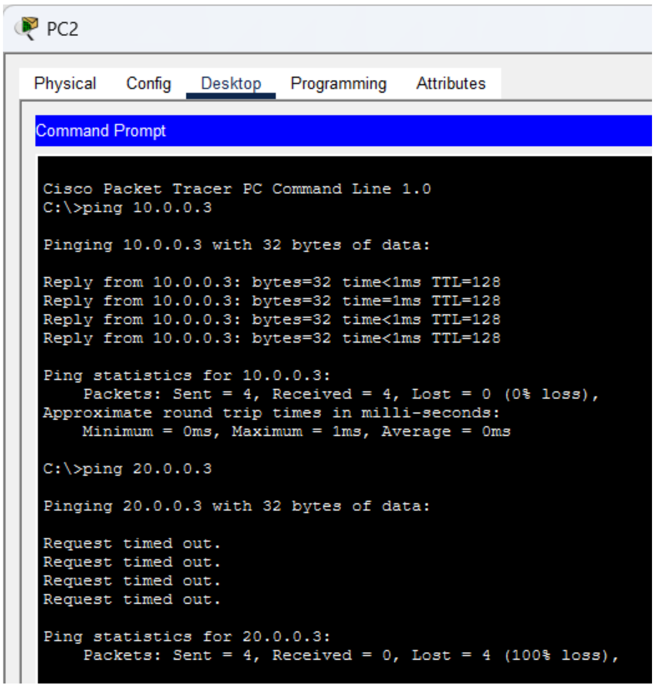
Result :-

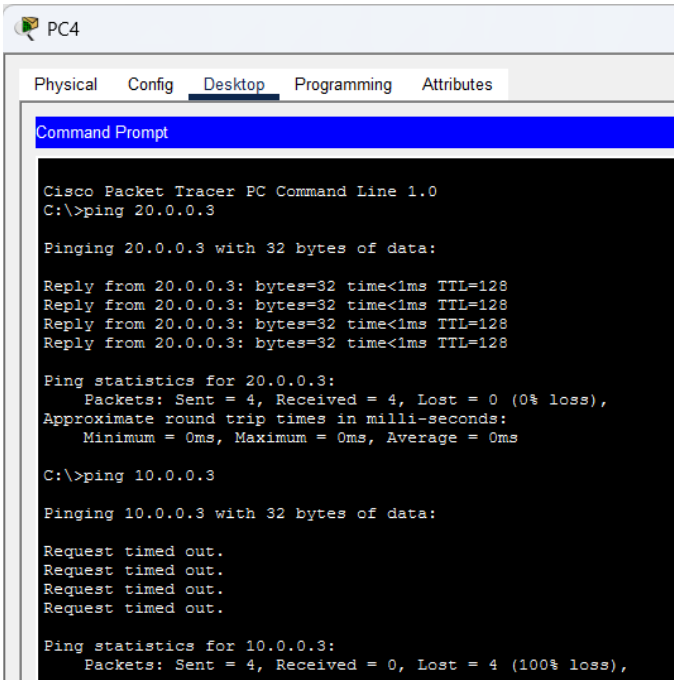


4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.



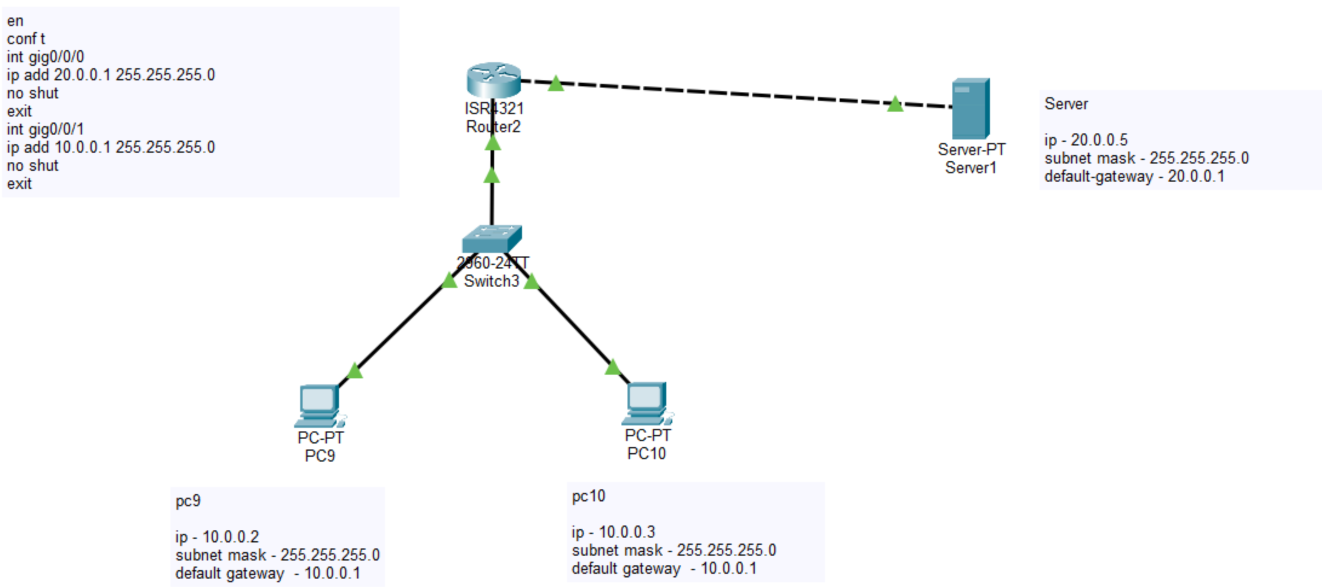
Result :-





Devices in the same VLAN communicated successfully, while devices in different VLANs could not communicate. This shows that VLANs separate the network into different groups.

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.



Step 2 :- NAT Configuration Steps :-

- Router(config)#int g0/0/1
- Router(config-if)#ip nat inside
- Router(config-if)#exit
- Router(config)#
- Router(config)#int g0/0/0
- Router(config-if)#ip nat outside
- Router(config-if)#exit
- Router(config)#access-list 1 permit 10.0.0.0 0.0.0.255
- Router(config)#ip nat inside source list 1 interface g0/0/0 overload
- Router(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.5

Step 3:- Server0 → **Services** → **HTTP**

- HTTP = **ON**

Step 4:- pc ----- web browser ---- http://20.0.0.5